THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL UNIT TEST SET 3

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Three coins are tossed. The probability of getting at most one head is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{2}{3}$

2. The value of the 20th term minus the 10th term of the AP $3, 10, 17, 24, \ldots$ is: 1

(a) 77 **(b)** 63 **(c)** 80 **(d)** 70

3. The length of a chord of a circle of radius 10 cm lying 8 cm from the centre is (in cm): 1

(a) 14 **(b)** 13 **(c)** 12 **(d)** 11

4. The length of the arc of a sector of angle 240° in a circle of radius 21 cm (take $\pi = \frac{22}{7}$) is (cm): 1

(a) 90 **(b)** 88 **(c)** 87 **(d)** 89

5. Two dice are rolled together. The probability that the sum is 9 is: 1

(a) $\frac{1}{10}$ **(b)** $\frac{2}{9}$ **(c)** $\frac{8}{9}$ **(d)** $\frac{1}{9}$

6. In an AP, $t_1 = 50$ and $t_9 = 26$. The common difference is: 1

(a) -4 **(b)** -2 **(c)** -3 **(d)** -1

7. A chord of length 16 cm lies 6 cm from the centre of a circle. The radius (in cm) is: 1

(a) 12 **(b)** 10 **(c)** 9 **(d)** 11

8. A triangle has area 84 cm^2 and one side 21 cm. The altitude on this side (cm) is: 1

(a) 10 **(b)** 7 **(c)** 9 **(d)** 8

9. Two dice are rolled together. The probability of getting a sum greater than 9 is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{1}{7}$ **(c)** $\frac{5}{6}$ **(d)** $\frac{1}{3}$

10. In an AP, $d = -2$ and $t_8 = 16$. The first term is: 1

(a) 30 **(b)** 31 **(c)** 32 **(d)** 29

11. In a circle with centre O, chord AB has $\angle AOB = 84^\circ$. Then $\angle OBA =$ 1

(a) 38° **(b)** 68° **(c)** 58° **(d)** 48°

12. The height of a parallelogram of area 184 cm^2 and base 23 cm is (cm): 1

(a) 9 **(b)** 8 **(c)** 7 **(d)** 10

13. A die is rolled. The probability of getting a prime number or a multiple of 3 is: 1

(a) $\frac{2}{3}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{3}{4}$

14. The 4th and 9th terms of an AP are 14 and 4. Its 15th term is: 1

(a) -9 **(b)** -8 **(c)** -7 **(d)** -6

15. The angles of a cyclic quadrilateral $ABCD$ at A and C are in the ratio $4:5$. Then $\angle A =$ 1

(a) 80° **(b)** 70° **(c)** 100° **(d)** 90°

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. The probability of winning a game is $\frac{2}{5}$. Find the probability of not winning, and the expected number of wins in 200 games. 2

17. The second and fifth terms of a GP are 6 and 162. Find the common ratio and the first term. 2

18. Find the length of the chord of a circle of radius 25 cm that is 15 cm from the centre. 2

19. A triangle has sides 13 cm, 20 cm and 21 cm. Find its area and the altitude on the side 21 cm. 2

20. Two dice are rolled together. Find the probability that the sum of the numbers is 8. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A die is rolled. Find the probability of getting 3
(i) a prime number,
(ii) a multiple of 3,
(iii) a prime number or a multiple of 3.

22. Find the number of terms of the AP $101, 94, 87, \ldots$ whose last term is -32, and its sum. 3

23. In a cyclic quadrilateral $ABCD$, $\angle A = (4x + 12)^\circ$ and $\angle C = (2x - 6)^\circ$. Find x, $\angle A$ and $\angle C$. 3

24. A rectangular sheet 25 cm $\times$ 28 cm has a semicircle of diameter 28 cm added on one breadth (take $\pi = \frac{22}{7}$). Find the perimeter and the area of the whole figure. 3

OR

In a survey of 300 people, 110 like tea only, 60 like coffee only and 40 like both. Find the probability that a person chosen at random likes tea (tea only or both), and the probability that the person likes neither (the rest). 3

25. The 2nd term of an AP is 26 and the 9th term is -2. Find the AP and its 14th term. 3

OR

O is the centre of a circle and $\angle AOB = 88^\circ$. A point C lies on the major arc and D on the minor arc. Find $\angle OAB$, $\angle ACB$ and $\angle ADB$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A chord AB of a circle with centre O is 24 cm long and the radius is 15 cm. Find the distance of the chord from O and the area of triangle OAB. 5

OR

A rectangular plot 50 m $\times$ 40 m has a circular pond of radius 14 m in it (take $\pi = \frac{22}{7}$). Find the area of the plot left, and the cost of turfing it at ₹4 per m^2. 5

27. Two dice are rolled together. Find the probability of 5

(i) a sum of 4,

(ii) a sum of 10,

(iii) a sum of 4 or 10.

OR

A ladder has 15 rungs. The bottom rung is 90 cm long and each rung above is 3 cm shorter than the one below. Find the length of the top rung and the total length of wood needed for the rungs. 5

END OF QUESTION PAPER **Total: 50 Marks**

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